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Assignment 4, Part I

We want to investigate whether the smell of fresh baking at a bakery has a causal effect on sales and design a *randomized controlled experiment* (or *trial*) as follows:

- over 10 randomly chosen days (*treatment group*) we open windows and observe the following sales U.S. dollars:
1216, 1221, 1223, 1224, 1234, 1238, 1252, 1252, 1258, 1260
- independently, over 12 randomly selected days (*control group*) we keep windows closed and observe the following sales in U.S. dollars:
1195, 1201, 1206, 1208, 1209, 1219, 1219, 1223, 1237, 1237, 1243, 1245

Assuming samples come from normal populations with unknown means μ_1 and μ_2 and same variance, test if there is a difference in expected sales. Answer in steps:

1. Compute the sample means of the treatment and control groups. What is the estimate of the expected difference, $\mu_1 - \mu_2$, in sales?
2. Test $H_0 : \mu_1 - \mu_2 = 0$ vs. $H_1 : \mu_1 - \mu_2 \neq 0$ (i.e., run a two-sided test) at the $\alpha = 0.05$ significance level. Report the p-value and your conclusion.
Hint: You need to use a t-test with pooled variance.

3. Alternatively, we assume the following relation:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

where y_i denotes sales on day i , x_i is a *dummy variable* that takes values 1 if windows were open on day i and 0 if closed, and $\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$, where σ^2 is constant (*homoscedasticity*).

Estimate the slope β_1 and then test $H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 \neq 0$ at the $\alpha = 0.05$ level. Is there evidence that $\beta_1 \neq 0$? Report the p-value and your conclusion.

4. Compare the p-values obtained in (2) and (3).
5. Discuss whether there is a causal effect of the smell of fresh baking on sales. Consider other ways windows could impact sales beyond the smell. How might one improve this experiment?

Assignment 4, Part 1

1) Compute sample means:

$$\text{treatment group} = \mu_1 = (1216 + 1221 + 1223 + 1224 + 1234 + 1238 + 1252 + 1252 + 1258 + 1260) / 10 = \boxed{1237.80}$$

$$\text{control group} = \mu_2 = (1195 + 1201 + 1206 + 1208 + 1209 + 1219 + 1223 + 1237 + 1237 + 1243 + 1245) / 12 = \boxed{1220.16}$$

$$\text{expected difference} = E[\mu_1 - \mu_2] = \boxed{17.63}$$

$$2) H_0 \rightarrow \mu_1 - \mu_2 = 0 \quad \bar{x}_1 = 1237.80 \quad n_1 = 10$$

$$H_1 \rightarrow \mu_1 - \mu_2 \neq 0 \quad \bar{x}_2 = 1220.16 \quad n_2 = 12$$

pooled variance

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(10 - 1)(276.18) + (12 - 1)(289.96)}{10 + 12 - 2} =$$

$$\frac{9(276.18) + 11(289.96)}{20} = 283.26 \quad s_p^2 = 283.76$$

$$s_p = \sqrt{283.76}$$

standard error

$$SE(\bar{x}_1 - \bar{x}_2) = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = 16.85 \sqrt{\frac{1}{10} + \frac{1}{12}} = 7.21$$

test statistic

$$TS = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{SE} = \frac{17.63}{7.21} = 2.45$$

$$\boxed{\text{p-value: } 0.024}$$

Conclusion:

The p-value is smaller than the alpha of 0.05, so we can reject the null hypothesis and say that windows open does seem to be associated with a positive effect on sales.

4) for dummy variables:

$$\hat{\beta}_0 = \bar{y}_2 \text{ (control)}$$

$$\hat{\beta}_1 = \bar{y}_1 - \bar{y}_2$$

treatment control

$$\boxed{\hat{\beta}_1 = 17.63 \leftarrow \text{slope}}$$

standard error

$$SE(\hat{\beta}_1) = s_p \sqrt{\frac{1}{10} + \frac{1}{12}} = 16.85 \sqrt{\frac{1}{10} + \frac{1}{12}} = 7.21$$

test statistic

$$TS = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} = \frac{17.63}{7.21} = 2.45$$

p-value

$$\boxed{P = 0.024}$$

conclusion

The p-value ends up being essentially the same and again, we can reject the null hypothesis and say that windows open seems to be associated with a positive effect on sales.

5) In the previous conclusions, we said that windows open was associated with a positive effect on sales but we cannot say there is a causal effect because other things that are associated with windows open could be the real "cause" and we don't have enough information about those. For example windows open could mean better moods due to good weather, more people out and about due to good weather, etc. One could improve the experiment by isolating these potential causes.